Fernando Fernández Andueza

Gameplay and AI Programmer

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Valencia, Spain. EU citizen, open to relocation and to remote work. Spanish (native), English (intermediate).

Profile

Game programmer focused on game AI, gameplay systems and procedural generation, working in C++ with Unreal Engine 5 and in C# with Unity. I have a Higher National Diploma (HND) in video game programming from ESAT and spent two and a half years as a fullstack developer before moving into games.

Technical skills

Engines and AI: Unreal Engine 5 (UE5) with C++ and Blueprints, Unity (C#), StateTree, Behavior Trees, EQS (Environment Query System), AI Perception, Animation Montages

Programming languages: C++ (C++17, C++20, C++23, STL), C#, C, ARM64 assembly, Lua

Graphics and systems: OpenGL 4.5 DSA, deferred shading, PBR, vertex and fragment shaders, Entity Component System (ECS), multithreading and job systems, memory management

Tools: Visual Studio debugger, Unreal Insights, RenderDoc, Git, Perforce, Plastic SCM

Outside games: TypeScript, JavaScript, HTML, CSS, Vue.js, SQL Server, Java, VB.NET

Experience

Gameplay Programmer, Oniric Tales — Dec 2024 – Jun 2025

Roadtrip: The Engine of Madness (Unity)

- Built the core architecture of the game.
- Built the AI with Unity's Behavior Graph, including a perception system written entirely by me.
- Built a component-based dialogue system and the car radio, which plays songs on several stations like a real car radio.
- Also worked on other gameplay mechanics, profiling and bug fixing.

Fullstack Developer, AHORA — Sep 2020 – Mar 2023

- Developed business management software with FlexyGo, the company's in-house low-code framework, using C#, VB.NET, SQL Server, JavaScript, jQuery, HTML and CSS.
- Worked on a production plant's back office, the forklift drivers' handheld app, quality-control tools and an offline mobile clock-in app with facial recognition. Built a ZPL label-printing app on my own.
- Also worked on apps for maintenance, HR, sales and real estate management. Handled client incidents, often on site, in a Scrum team.

Selected projects

Beerserker: The Stolen Brew — Unreal Engine 5, C++

Final-year project at ESAT with a team of about 20, co-developed with Grumpy Games. Free, in Early Access on Steam.

- Built the four enemy types (melee, ranged, explosive and a tank mini-boss) in StateTree, with custom tasks and conditions in C++ and Blueprints.
- Wrote an object pool for enemies, projectiles, particles and decals, packaged as a plugin so other projects can reuse it.
- The ranged enemy uses EQS to find somewhere to retreat to, based on distance to the player and line of sight.
- The tank favours the melee attacks it has used least, so it doesn't keep repeating the same one.
- Built the enemy spawners, set up and uploaded the Steam builds, and took part in designing the architecture of a custom action system based on GAS (Gameplay Ability System).

XEMA (custom C++ engine) — C++, OpenGL 4.5

- Includes an ECS, deferred PBR rendering, shadow mapping, instancing, a job system that loads meshes asynchronously, Lua scripting and PhysX.
- Ported to Nintendo Switch (tested only on an emulator). PS4 port in progress: so far the ECS and basic geometry run on a devkit.

Education and certifications

HND in Video Game Programming, ESAT (Escuela Superior de Arte y Tecnología), Valencia — 2023–2026

Specialised in C++. Built my own engine and implemented Wave Function Collapse, cellular automata, L-systems, a templated FSM and A* pathfinding. Learned ARM64 assembly to find and fix bottlenecks in inner loops.

DAM (Multiplatform Application Development), IES El Grao, Valencia — 2018–2020

Java for desktop and Android, relational databases, HTML and CSS, OOP fundamentals and version control.

Certified Scrum Master and Product Owner, Scrum.org / International Agile Institutes